Slop / University

Locker Ping

Design and a Six-Week Field Deployment of an Automated Collection-Reminder System for the Campus Parcel-Locker Bank

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Background

Facilities holds the campus parcel-locker bank's foyer stock to a 24-hour collection-speed target, covered by manual courtesy calls that reach a fraction of overdue collections. Could a rule-based reminder engine reach more recipients before the window closes, without adding staff hours?

Aims

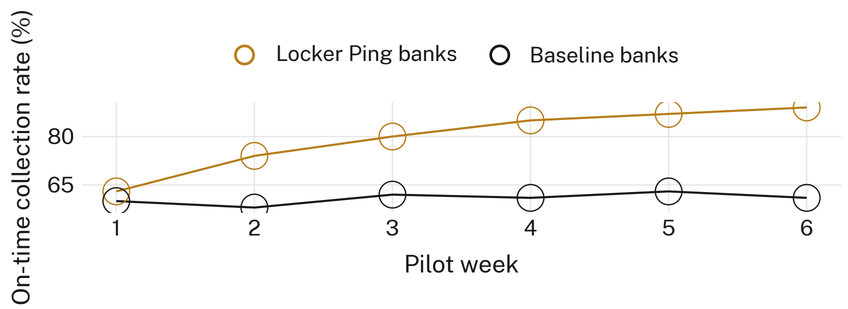
- Design a rule-based engine that fires a reminder as a locker's 24-hour window nears expiry.
- Deploy the engine across a subset of banks alongside business-as-usual sites.
- Evaluate delivery, recipient response, and on-time collection rate over a field pilot.

Methods

- Push notification at T-6h, SMS at T-1h · single-reply opt-out · deployed to 6 of 14 campus locker banks
- 8 banks continued the manual courtesy-call baseline, unchanged
- 6-week pilot, Apr-May 2026 · RFID drop/collection timestamps logged at all 14 banks

Results

Nudged banks reach 89% on-time collection by pilot end, up from a 61% baseline mean across the un-nudged sites; the gap opens within the first fortnight and holds across all six weeks.



Weekly on-time collection rate, Locker Ping banks (n = 6) vs baseline banks (n = 8), 21 Apr-30 May 2026: nudged sites reach 89% by week 6 against a stable 58-63% baseline band.

Discussion & conclusions

The engine moves recipients, not shelf capacity: nudged banks report no change in average dwell time per parcel, and overdue collections concentrate at the same handful of oversubscribed lockers regardless of reminder delivery. Next: pair the reminder with a capacity-aware routing rule before the next expansion round.

References

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De-identified RFID drop/collection timestamps and delivery logs only; opt-outs excluded from all counts; thresholds pre-registered.

"The nudge doesn't check whether there's room."

89% of nudged parcels cleared before the 24-hour deadline, up from a 61% baseline mean, over a six-week pilot (Apr-May 2026).